Little Lab Coats

👐 The Great Push & Pull Sort 👐

Physical Science — Unit 1, Lesson 1

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Forces

Everything that moves gets a push or a pull — can you figure out which is which?

📋 **NGSS Standard:**
K-PS2-1 — Plan and conduct an investigation to compare the effects of different strengths or different directions of pushes and pulls on the motion of an object
K-PS2-2 — Analyze data to determine if a design solution works as intended

🎯 Learning Objective

Students will investigate the Great Push & Pull Sort, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

Things move because something **pushes** them or **pulls** them! A push moves something *away* from you. A pull moves something *toward* you. In this lesson, your kids will explore real objects and everyday actions, then sort them into push and pull categories.

This is a preview. Unlock the full lesson.

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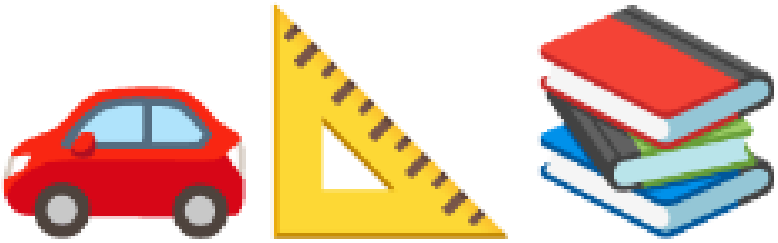
🏎️ Ramp Racers 🏎️

Unit 1, Lesson 2 — Forces & Motion: How Far Will It Go?

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Forces & Motion



Higher ramp = more speed = farther roll! Let's find out!

📋 **NGSS Standard:**
K-PS2-1 — Plan and conduct an investigation to compare the effects of different strengths or different directions of pushes and pulls on the motion of an object.

🎯 Learning Objective

Students will investigate ramp Racers, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

When you let a toy car roll down a ramp, **gravity pulls it down**. A higher ramp means gravity pulls the car for longer, so it goes **faster** and rolls **farther**! In this experiment, you'll build ramps at different heights and discover how changing height changes how far things move.

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💪 Strong Push, Gentle Push 👊

Unit 1, Lesson 3 — How Hard You Push Changes How Far Things Go!

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Forces

A strong push sends things flying — a gentle push barely moves them!

📋 **NGSS Standard:**
K-PS2-1 — Plan and conduct an investigation to compare the effects of different strengths or different directions of pushes and pulls on the motion of an object.

🎯 Learning Objective

Students will investigate strong Push, Gentle Push, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

A **push** is a force that moves things away from you. But not all pushes are the same. A **gentle push** moves an object a little bit. A **strong push** moves it *a lot*. Today, we're going to test this with our own hands.

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🧲 Magnet Magic 🧲

Physical Science — Unit 1, Lesson 4

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Forces 🎒 **Kit Lesson**

🎒 **Kit Lesson** — Specialty materials included in the physical kit

Invisible pushes and pulls — discover the magic of magnets!

📋 **NGSS Standard:**
K-PS2-1 — Plan and conduct an investigation to compare the effects of different strengths or different directions of pushes and pulls on the motion of an object
K-PS2-2 — Analyze data to determine if a design solution works as intended

🎯 Learning Objective

Students will investigate magnet Magic, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

You've learned that pushes and pulls make things move — but did you know some forces can work **without touching**? Magnets have a special power: they can **pull** some materials toward them without touching them. In this lesson, your kids will explore which materials are magnetic and which aren't, discovering that not all pushes and pulls require contact.

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💨 Wind Wagon Engineering Challenge 💨

Physical Science + Engineering — Unit 1, Lesson 5

📚 **Grade:** Kindergarten ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Physical Science — Forces + Engineering 🎒 **Kit Lesson**

🎒 **Kit Lesson** — Specialty materials included in the physical kit

Can you design a wagon that moves using only the wind? Engineer it, test it, make it better!

📋 **NGSS Standard:**
K-2-ETS1-1 — Ask questions, make observations, and gather information about a situation people want to change to define a simple problem that can be solved through the development of a new or improved object or tool
K-2-ETS1-2 — Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem
K-2-ETS1-3 — Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs

🎯 Learning Objective

Students will investigate wind Wagon Engineering Challenge, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🌟 The Big Idea

Wind is just moving air — and moving air can push! Engineers use the wind's push to move sailboats, power wind turbines, and even fly kites. In this challenge, your kids will become engineers: they'll design, build, test, and improve a **wind wagon** — a wheeled vehicle that moves when pushed by wind from a fan (or your breath). This is a full engineering design challenge using the Engineering Design Process: Ask, Imagine, Plan, Create, Test, and Improve!

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🎢 Marble Run Madness 🎢

Physical Science — Unit 1, Lesson 6 (Capstone)

📚 **Grade:** Kindergarten ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Physical Science — Forces 🏆 **Capstone Lesson**

🏆 **Capstone Lesson** — This final lesson brings together all concepts from the unit. Students apply everything they've learned.

Put all your push & pull knowledge together — build the ultimate marble run!

📋 **NGSS Standard:**
K-PS2-1 — Plan and conduct an investigation to compare the effects of different strengths or different directions of pushes and pulls on the motion of an object
K-PS2-2 — Analyze data to determine if a design solution works as intended

🎯 Learning Objective

Students will investigate marble Run Madness, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

This is it — the grand finale of Unit 1. Your kids have learned about pushes, pulls, forces, magnets, and wind. Now they'll apply *everything* they've learned by building a **marble** [illegible] ble down and the track pushes it along the path. They'll explore how the angle of a ramp changes the marble's speed, how curves affect direction, and how to solve problems when the marble gets stuck. It's hands-on physics at its finest.

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